

MD. AKASH MIAH

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PROFESSIONAL SUMMARY

AI Engineer with experience building machine learning, computer vision, data retrieval, and AI applications. Skilled in Python, SQL, PyTorch, Scikit-learn, PostgreSQL, FastAPI, and Docker. Experienced in model training, evaluation, RAG pipelines, APIs, and deployment, and interested in Data Science, Machine Learning, and Data Engineering roles.

TECHNICAL SKILLS

Programming & Querying	Python, SQL, Go, TypeScript
Data Science & ML	PyTorch, TensorFlow, Scikit-learn, Transformers, OpenCV, YOLOv8, RT-DETR, Model Training, Model Evaluation, Fine-tuning
Data & Retrieval	PostgreSQL, Redis, Qdrant, PGVector, Pinecone, ChromaDB, Semantic Search, RAG Pipelines
AI & LLM Systems	LLMs, AI Agents, Prompt Engineering, Function Calling, Tool Use, Model Evaluation
LLM Frameworks	LangChain, LangGraph, LlamaIndex
Backend & APIs	FastAPI, REST API Development, Webhooks, JWT Authentication
MLOps & Deployment	Docker, Linux, Git, vLLM, Model Serving, Monitoring, Rate Limiting, CI/CD
AI Tools & Providers	OpenAI Codex, OpenAI API, Claude API, Gemini, Groq, Claude Code (certification), Self-Hosted Models

EDUCATION & PROFESSIONAL DEVELOPMENT

B.Sc. in Computer Science and Engineering	2025
North South University	Dhaka, Bangladesh
Short Course: Data Science — Prof. Jennifer Widom, Stanford University	Feb 2024
Anthropic Certifications: Building with the Claude API · Claude Code in Action · Introduction to Agent Skills · Introduction to Model Context Protocol	May 2026

PROFESSIONAL EXPERIENCE

AI Engineer (AI Systems)	Jul 2025 – Present
Betopia LTD	Dhaka, Bangladesh
- Contribute to Betopia’s production AI platform and AI-agent workflows across model integration, data retrieval, API development, deployment, and reliability improvements. - Built and deployed AI applications using OpenAI, Claude, Gemini, and self-hosted LLMs through a unified model-access layer. - Developed RAG and semantic retrieval pipelines using Qdrant and PGVector to support knowledge retrieval and contextual responses. - Developed AI agents and coding assistants with LangChain-based workflows for engineering and operational task automation. - Contributed to a centralized LLM platform that provides unified API access to commercial and open-source model providers, including routing and provider failover. - Built FastAPI backend services and inference APIs with PostgreSQL, Redis, and JWT authentication. - Containerized AI services with Docker and contributed to deployment workflows for production environments. - Improved inference performance through model-serving optimizations, intelligent routing, and infrastructure enhancements. - Fine-tuned transformer-based models for domain-specific use cases using proprietary datasets. - Collaborated with engineering and business stakeholders to translate requirements into practical data and AI-powered services.	

SELECTED PROJECTS

Adder — Desktop Data Science Platform

2026–Present

React, TypeScript, Tauri, Rust, Python, SQLite

- Building a cross-platform desktop platform for dataset exploration, visualization, preparation, model training, AutoML, and notebook-based analysis.
- Developing a Git-like dataset lifecycle with cleaning pipelines, immutable versions, commit messages, lineage graphs, and version-specific change tracking.
- Creating native interactive charting, ML diagnostics, and Python notebook execution workflows with streamed outputs and project-aware data integration.

Betopia AI Ecosystem

Go, Python, TypeScript, FastAPI, vLLM, PostgreSQL, Redis

- Worked as the AI Systems Engineer on a production AI and inference ecosystem covering conversational AI, coding agents, and developer-facing APIs.
- Integrated OpenAI, Claude, Gemini, Groq, and self-hosted models through a unified orchestration layer.
- Built multi-agent workflows for code generation, task execution, tool use, and workflow automation.
- Implemented RAG pipelines with semantic search and vector retrieval capabilities.
- Developed FastAPI and Go services supporting LLM inference and data-intensive AI application workloads.
- Contributed to authentication, rate limiting, monitoring, RBAC, and provider failover mechanisms for reliable production operation.

Automated Grocery Checkout System

Python, OpenCV, YOLO, RT-DETR

- Built a vision-driven checkout framework for multi-product detection, GTIN-based identification, database retrieval, and human-in-the-loop POS billing.
- Prepared a custom ten-product dataset across varied lighting, backgrounds, orientations, and viewing distances, including 154 background-only scenes for robustness testing.
- Compared YOLOv8m, YOLOv11m, YOLOv12m, and RT-DETR across accuracy, localization, latency, model size, and background false positives; all achieved **99.4–99.5% mAP@0.5**.
- Achieved **93.5% mAP@0.5:0.95** with YOLOv8m and two background false positives; YOLOv11m delivered **544.8 ms/image** inference and a **38.6 MB** model size.

LUMINA / Auralis — AI-Powered Medical Reporting & Writing Platform

- Developed a cross-platform AI medical reporting and writing platform with desktop, web, and admin tools.
- Built AI workflows for proofreading, summarization, clinical interpretation, and medical diagram generation.
- Engineered REST APIs with PostgreSQL, authentication, and subscription management.
- Created a Rust desktop assistant with global hotkeys and multilingual support.

Technologies: Rust, Python, Node.js, PostgreSQL, React, OpenAI API, Docker